Quasi-Newton Methods

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Notations. We write $\langle \cdot, \cdot \rangle$ for the Euclidean inner product and $\| \cdot \|$ for its induced norm. Let $\mathbb{S}^n$, $\mathbb{S}^n_+$, and $\mathbb{S}^n_{++}$ denote the real symmetric, positive semidefinite, and positive definite $n \times n$ matrices, respectively. For $A, B \in \mathbb{S}^n$, we write $A \succeq B$ when $A - B \in \mathbb{S}^n_+$.

1 Preliminaries

1.1 Useful inequalities

Lemma 1.1 (Arithmetic–geometric mean inequality)**.** *For any integer $K \geq 1$ and nonnegative scalars $r_1, \ldots, r_K$,*

$$\left(\prod_{k=1}^K r_k\right)^{1/K} \leq \frac{1}{K} \sum_{k=1}^K r_k.$$

Lemma 1.2. *If $A \in \mathbb{S}^n_{++}$ and $\mu I \preceq A \preceq LI$ for $0 \leq \mu \leq L$, then, for every $x \in \mathbb{R}^n$,*

$$\frac{\mu}{2}\|x\|^2 \leq \frac{1}{2}\langle x, Ax \rangle \leq \frac{L}{2}\|x\|^2.$$

1.2 Convexity and smoothness

Convexity. A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if

$$f\big(\theta x + (1-\theta)y\big) \leq \theta f(x) + (1-\theta) f(y)$$

for all $x, y \in \mathbb{R}^n$ and $\theta \in [0, 1]$. A useful consequence for differentiable convex functions is the first-order inequality.

Proposition 1.1. *If $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable and convex, then*

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle \qquad \textit{for all } x, y \in \mathbb{R}^n.$$

Smoothness. A continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is L-smooth if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \qquad \text{for all } x, y \in \mathbb{R}^n.$$

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Lemma 1.3 (Descent lemma)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is L-smooth, then*

$$f(x) \le f(y) + \langle \nabla f(y), x - y \rangle + \tfrac{L}{2}\|x - y\|^2 \qquad \textit{for all } x, y \in \mathbb{R}^n.$$

Descent direction I. For a differentiable function f, a *descent direction* at x is a vector d satisfying

$$\langle d, \nabla f(x) \rangle < 0.$$

For fixed d, first-order expansion gives, as $\varepsilon \downarrow 0$,

$$f(x + \varepsilon d) = f(x) + \varepsilon \langle d, \nabla f(x) \rangle + o(\varepsilon).$$

Hence $f(x + \varepsilon d) < f(x)$ for all sufficiently small $\varepsilon > 0$. If f is L-smooth with $L > 0$, the descent lemma gives the explicit bound

$$f(x + \varepsilon d) \le f(x) + \varepsilon \langle d, \nabla f(x) \rangle + \tfrac{L\varepsilon^2}{2}\|d\|^2.$$

Taking $\varepsilon = -\langle d, \nabla f(x) \rangle / (L\|d\|^2) > 0$ yields

$$f(x + \varepsilon d) \le f(x) - \tfrac{\langle d, \nabla f(x) \rangle^2}{2L\|d\|^2}.$$

In particular, if $\nabla f(x) \neq 0$ and $d = -\nabla f(x)$, then $\varepsilon = 1/L$ and $f\big(x - \frac{1}{L}\nabla f(x)\big) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$.

Descent direction II. Let f be differentiable and convex, and let $x^\star$ be a minimizer. Consider the squared distance to this minimizer,

$$\delta(x) := \tfrac{1}{2}\|x - x^\star\|^2.$$

The function δ is convex, 1-smooth, and 1-strongly convex, but its gradient $\nabla\delta(x) = x - x^\star$ depends on the unknown minimizer. Nevertheless, convexity of f gives

$$\langle -\nabla f(x), \nabla\delta(x) \rangle \le f(x^\star) - f(x) < 0 \qquad \text{whenever } f(x) > f(x^\star).$$

Thus $-\nabla f(x)$ is also a descent direction for δ at every nonoptimal x. Indeed, for $g = \nabla f(x)$,

$$\delta(x - \eta g) = \delta(x) - \eta \langle g, x - x^\star \rangle + \tfrac{\eta^2}{2}\|g\|^2,$$

so a sufficiently small $\eta > 0$ reduces the distance to $x^\star$.

2 Algorithms

2.1 Line search with a scaled gradient direction

Suppose f is L-smooth and μ-strongly convex, with $L \ge \mu > 0$. Given a symmetric positive definite matrix H satisfying $\lambda_{\min} I \preceq H \preceq \lambda_{\max} I$, where $0 < \lambda_{\min} \le \lambda_{\max}$, consider the scaled gradient step

$$d = -H\nabla f(x), \qquad x^+ = x + \hat{t} d.$$

Write $\|v\|_H^2 = \langle v, Hv \rangle$. At a nonstationary point, d is a descent direction because

$$\langle d, \nabla f(x) \rangle = -\|\nabla f(x)\|_H^2 < 0.$$

Backtracking line search starts from a trial step $t_0 > 0$ and repeatedly multiplies it by $\beta \in (0,1)$ until the Armijo condition

$$f(x + \hat{t}d) - f(x) \le c\hat{t}\langle d, \nabla f(x)\rangle = -c\hat{t}\|\nabla f(x)\|_H^2$$

holds, where $c \in (0,1)$. Every $0 < t \le t^\star$, with $t^\star = 2(1-c)/(L\lambda_{\max})$, satisfies this condition. Indeed, the descent lemma and $H^2 \preceq \lambda_{\max} H$ give

$$\begin{aligned} f(x+td) &\le f(x) + t\langle \nabla f(x), d\rangle + \tfrac{Lt^2}{2}\|d\|^2 \\ &= f(x) - t\|\nabla f(x)\|_H^2 + \tfrac{Lt^2}{2}\|H\nabla f(x)\|^2 \\ &\le f(x) - t\|\nabla f(x)\|_H^2 + \tfrac{L\lambda_{\max} t^2}{2}\|\nabla f(x)\|_H^2 \\ &= f(x) - t\big(1 - \tfrac{L\lambda_{\max} t}{2}\big)\|\nabla f(x)\|_H^2 \\ &\le f(x) - ct\|\nabla f(x)\|_H^2. \end{aligned}$$

Thus backtracking terminates and the accepted step satisfies $\hat{t} \ge \min\{t_0, \beta t^\star\}$. If $t_0 \ge t^\star$, this reduces to $\hat{t} \ge \beta t^\star$. For example, taking $c = 1/2$ gives

$$\begin{aligned} f(x^+) - f(x) &\le -\tfrac{\hat{t}}{2}\|\nabla f(x)\|_H^2 \\ &\le -\tfrac{\lambda_{\min}}{2}\min\big\{t_0, \tfrac{\beta}{L\lambda_{\max}}\big\}\|\nabla f(x)\|^2. \end{aligned}$$

2.2 Proximal interpretation of gradient descent

For a differentiable f and $\eta > 0$, define its first-order model at x by $\ell_x(z) = f(x) + \langle \nabla f(x), z - x\rangle$. The gradient step minimizes this model plus a quadratic penalty:

$$x^+_{\mathrm{GD}} = x - \eta\nabla f(x) = \arg\min_z \big\{\ell_x(z) + \tfrac{1}{2\eta}\|z - x\|^2\big\}.$$

For a proper, closed, convex function f, the proximal point step instead minimizes the original function plus the same penalty:

$$x^+_{\mathrm{Prox}} = \arg\min_z \big\{f(z) + \tfrac{1}{2\eta}\|z - x\|^2\big\}.$$

2.3 Newton's method

Consider the unconstrained minimization problem

$$\min_{x\in\mathbb{R}^n} f(x).$$

Suppose f is twice continuously differentiable and $x^\star$ is a minimizer. On a convex neighborhood of $x^\star$, assume $\nabla^2 f(z) \succeq \mu I$ for $\mu > 0$, and that the Hessian is M-Lipschitz for a constant $M > 0$:

$$\|\nabla^2 f(u) - \nabla^2 f(v)\| \le M\|u - v\|.$$

Here the matrix norm is the operator norm induced by the Euclidean norm. Newton's method takes the step

$$x^+ = x - [\nabla^2 f(x)]^{-1}\nabla f(x).$$

Local quadratic convergence. Since $\nabla f(x^\star) = 0$, the fundamental theorem of calculus gives

$$\nabla f(x) = \int_0^1 \nabla^2 f\big(x^\star + s(x - x^\star)\big)(x - x^\star)\, ds.$$

Consequently,

$$\begin{aligned} x^+ - x^\star &= x - x^\star - [\nabla^2 f(x)]^{-1} \nabla f(x) \\ &= [\nabla^2 f(x)]^{-1} \int_0^1 \big[\nabla^2 f(x) - \nabla^2 f\big(x^\star + s(x - x^\star)\big)\big](x - x^\star)\, ds. \end{aligned}$$

The inverse Hessian has norm at most $1/\mu$, so

$$\begin{aligned} \|x^+ - x^\star\| &\le \tfrac{M}{\mu} \int_0^1 (1 - s)\|x - x^\star\|^2\, ds \\ &= \tfrac{M}{2\mu}\|x - x^\star\|^2. \end{aligned}$$

Thus Newton's method converges quadratically when started sufficiently close to $x^\star$.

Globalization. The bound contracts the error whenever $\|x - x^\star\| < 2\mu/M$, provided the iterates remain in the neighborhood where the assumptions hold. Newton's method can diverge from a distant starting point. To obtain global convergence, suppose additionally that $\mu I \preceq \nabla^2 f(x) \preceq LI$ for all $x \in \mathbb{R}^n$. The inverse Hessian then satisfies $L^{-1} I \preceq [\nabla^2 f(x)]^{-1} \preceq \mu^{-1} I$. Applying the line search of Section 2.1 with this scaled gradient direction and fixed t_0, β, and c gives convergence to $x^\star$ from any starting point.

2.4 Quasi-Newton methods

Newton's method requires computing $[\nabla^2 f(x)]^{-1} \nabla f(x)$ at each iteration, usually by solving a linear system. This can be computationally prohibitive in practice. The idea of quasi-Newton methods is:

- **Approximation:** replace $[\nabla^2 f(x)]^{-1}$ by an approximation matrix P.
- **Learning:** when P works poorly, update it to make it behave like $[\nabla^2 f(x)]^{-1}$.
- **Amortization:** amortize the cost of computing $[\nabla^2 f(x)]^{-1}$ across iterations.

There are many variants of quasi-Newton methods, based on different learning principles. We will cover them in class. Below we demonstrate three types. For simplicity, we focus on the quadratic minimization problem

$$\min_{x \in \mathbb{R}^n} f(x) := \tfrac{1}{2}\langle x, Ax\rangle - \langle b, x\rangle,$$

where $A \in \mathbb{S}^n_{++}$ and $\mu I \preceq A \preceq LI$ for $0 < \mu \le L$. Quadratic minimization is often a good surrogate for the local geometry of general unconstrained optimization problems. Indeed, if f is twice continuously differentiable near a minimizer $x^\star$, then

$$f(x) = f(x^\star) + \tfrac{1}{2}\langle x - x^\star, \nabla^2 f(x^\star)(x - x^\star)\rangle + o(\|x - x^\star\|^2).$$

In each template below, stop if $\nabla f(x) = 0$.

The following algorithms give one iteration of each template. In the two quasi-Newton variants, dist is a matrix distance or divergence chosen so that the constrained minimizer exists. For the quadratic problem, $y = As$, so $\langle s, y\rangle > 0$ whenever $s \neq 0$.

Algorithm 1 Quasi-Newton method in Hessian space (one iteration)

Require: Current iterate x and Hessian approximation $B \in \mathbb{S}^n_{++}$.

1: Choose $t > 0$ by backtracking line search along $-B^{-1}\nabla f(x)$.
2: $x^+ = x - tB^{-1}\nabla f(x)$.
3: $s = x^+ - x$, $y = \nabla f(x^+) - \nabla f(x)$.
4: $B_+ = \arg\min_{\substack{C \in \mathbb{S}^n_{++} \\ Cs = y}} \operatorname{dist}(C, B)$.

Algorithm 2 Quasi-Newton method in inverse Hessian space (one iteration)

Require: Current iterate x and inverse Hessian approximation $P \in \mathbb{S}^n_{++}$.

1: Choose $t > 0$ by backtracking line search along $-P\nabla f(x)$.
2: $x^+ = x - tP\nabla f(x)$.
3: $s = x^+ - x$, $y = \nabla f(x^+) - \nabla f(x)$.
4: $P_+ = \arg\min_{\substack{Q \in \mathbb{S}^n_{++} \\ Qy = s}} \operatorname{dist}(Q, P)$.

Algorithm 3 Space dilation (one iteration)

Require: Current iterate x and matrix P.

1: $x^+ = x - P\nabla f(x)$.
2: $v = \nabla f(x)/\|\nabla f(x)\|$, $r = (I - AP)v$.
3: **if** $r = 0$ **then**
4: $\quad P_+ = P$.
5: **else**
6: $\quad u = \frac{r}{\|r\|}$, $\eta = \frac{\langle Au, r\rangle}{\|Au\|^2}$.
7: $\quad P_+ = P + \eta uv^\top$.
8: **end if**

Algorithm 4 Hypergradient descent (one iteration)

Require: Current iterate x, matrix P, and learning rate $\eta > 0$.

1: $x^+ = x - P\nabla f(x)$.
2: $v = \nabla f(x)$, $u = -\nabla f(x^+)$.
3: $P_+ = P - \eta uv^\top$.